

SOFTWARE REQUIREMENT SPECIFICATION

FOR

HOSTEL MANAGEMENT

VERSION 1.0 APPROVED

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REVISION HISTORY

NAME	DATE	DESCRIPTION	VERSION
Group	06/06/2026	Initial Creation of the SRS Document	1.0

1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) document describes the requirements of the **Aruvi Hostel Management System (AHMS)**. The purpose of this document is to clearly define the functional and non-functional requirements of the system so that developers, evaluators, and stakeholders understand the system behavior and constraints.

1.2 Document Conventions

The following conventions are used in this document:

- **SHALL** indicates a mandatory requirement.
 - **SHOULD** indicates a recommended requirement.
 - **MAY** indicates an optional requirement.
 - All requirements are described in clear and unambiguous language.
-

1.3 Intended Audience and Reading Suggestions

This document is intended for:

- Faculty evaluators
- Software developers
- System testers
- Hostel administrators

Readers are advised to read the Overall Description section first to understand the system, followed by the System Features section for detailed requirements.

1.4 Product Scope

The Aruvi Hostel Management System is a web-based application designed to automate hostel operations such as student applications, priority-based ranking, hostel allocation, administration, and communication. The system aims to ensure transparency, efficiency, and fairness in hostel admission and management.

1.5 References

- IEEE 830 / IEEE 29148 Software Requirements Specification Standard
 - PHP Documentation
 - MySQL Documentation
-

2. Overall Description

2.1 Product Perspective

The Aruvi Hostel Management System is a standalone web application that replaces the existing manual and semi-manual hostel management processes. It operates as a centralized system with role-based access.

2.2 Product Functions

The major functions of the system include:

- User authentication and authorization
 - Hostel application submission
 - Application window and edit window control
 - Distance calculation from student residence
 - Priority score calculation
 - Rank list generation
 - Hostel allotment approval
 - Dashboards for Admin, Warden, and Student
 - Complaint and notification management
-

2.3 User Classes and Characteristics

Admin

- Has complete control over the system
- Calculates priority scores and publishes rank lists
- Manages applications, analytics, notifications, and approvals

Warden

- Manages allotted students
- Views inmate list
- Handles hostel-level complaints

Student

- Submits hostel application
 - Views application and allotment status
 - Raises complaints and receives notifications
-

2.4 Operating Environment

- Backend: PHP
 - Database: MySQL
 - Server: Apache (XAMPP)
 - Client: Web browser (Chrome, Firefox, Edge)
-

2.5 Design and Implementation Constraints

- The system must use PHP and MySQL
 - Role-based access must be enforced
 - Internet connectivity is required
 - Data accuracy depends on user input
-

2.6 User Documentation

The system will provide:

- Online help messages
 - Alert-based error handling
 - Basic user guidance through the interface
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2.7 Assumptions and Dependencies

- Students provide valid information

- Admin correctly sets application and edit windows
 - Server availability during usage
-

3. External Interface Requirements

3.1 User Interfaces

- Web-based responsive user interface
 - Common dashboard layout for all roles
 - Sidebar navigation and card-based design
-

3.2 Hardware Interfaces

No specific hardware interfaces are required.

3.3 Software Interfaces

- MySQL database
 - PHP session management
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3.4 Communications Interfaces

- HTTP/HTTPS protocol
-

4. System Features

4.1 User Authentication

Description:

The system authenticates users using User ID and password and redirects them to role-specific dashboards.

Functional Requirements:

- The system SHALL validate login credentials.
 - The system SHALL restrict access based on role.
 - The system SHALL display errors using alert messages.
-

4.2 Hostel Application Management

Description:

Students can submit hostel applications during the application window.

Functional Requirements:

- The system SHALL allow application submission only within the application window.
 - The system SHALL prevent editing after submission.
 - The system SHALL allow editing only during the edit window.
-

4.3 Priority Score Calculation and Rank List

Description:

Priority scores are calculated based on distance, family income, and PWD status.

Functional Requirements:

- The system SHALL calculate scores only after windows close.
 - The system SHALL generate a rank list based on scores.
 - The rank list SHALL be visible only to Admin.
-

4.4 Dashboards

Description:

The system provides dashboards based on user roles.

Functional Requirements:

- Admin dashboard SHALL provide analytics and approval options.
 - Warden dashboard SHALL show allotted students only.
 - Student dashboard SHALL show application status.
-

5. Other Non-Functional Requirements

5.1 Performance Requirements

The system SHALL support multiple concurrent users without performance degradation.

5.2 Security Requirements

- Passwords SHALL be stored using hashing techniques.
 - Unauthorized access SHALL be prevented.
-

5.3 Software Quality Attributes

- Usability
 - Reliability
 - Maintainability
 - Scalability
-

5.4 Business Rules

- Only Admin can calculate and publish rank list.
 - Score calculation is allowed only after edit window closure.
-

6. Other Requirements

- Data backup facility
 - Audit logging for admin actions
-

Appendix A: Glossary

- **AHMS** – Aruvi Hostel Management System
 - **PWD** – Persons With Disability
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Appendix B: To Be Determined List

- Mobile application support
- SMS and Email notifications